Chapter 6: Elementary Matrices Worksheet

1. Are each of the following statements true or false?

(a) For a 3×3 matrix A, replacing r_2 with $3r_2 + r_1$ is an elementary row operation.

(b) If A is a 3×3 matrix, and E_1 and E_2 are 3×3 elementary matrices, then $E_2E_1A = E_1E_2A$.

(c) If A, B are 4×4 matrices each with rank 4, then AB will also have rank 4.

2. Given $A = \begin{bmatrix} 0 & 0 & 2 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$:

(a) Compute A^{-1}.

(b) Use your answer to part (a) to find B if $A(B + I_3) = D$, where $D = \begin{bmatrix} 4 & 0 & 6 \\ 2 & 0 & 10 \\ 1 & 2 & 0 \end{bmatrix}$

3. Suppose A is a 4×4 that reduces to B via elementary row operations, where $B = \begin{bmatrix} 1 & 1 & 1 & -1 \\ 0 & 1 & 2 & -4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. Answer the following questions about A, if possible. Justify your answers.

(a) What is the rank of A?

(b) Is A invertible?

(c) What is the RREF of A?

(d) Does the homogeneous system $A\mathbf{x} = \mathbf{0}$ have (i) no solution, (ii) a unique solution, or (iii) infinitely many solutions?

(e) Does the system $A\mathbf{x} = \begin{bmatrix} 1 \\ -2 \\ 3 \\ 1 \end{bmatrix}$ have (i) no solution, (ii) a unique solution, or (iii) infinitely many solutions?